

diesel-electric drive system with electric double-layer capacitors.

Since 2003, Mannheim Stadtbahn in Mannheim, Germany, has been operating a light-rail vehicle which uses electric double-layer capacitors to store braking energy.

The transportation systems division of Siemens AG is developing a mobile energy storage based on double-layer capacitors, called Sibac Energy Storage, and also Sitrans SES—a stationary version integrated into the trackside power supply.

For more information, you may refer to:

1. Website 'www.act.jp/eng/index.htm'
2. Website 'http://fpfreflib.findlay.co.uk/articles/6610/if-the-cap-fits.pdf'
3. "Researchers Fired Up Over New Battery," Deborah Halber, MIT News Office, February 8, 2006
4. 'The Charge of the Ultra-capacitors,' IEEE Spectrum, November 2007

Q2. How can I construct a cathode ray oscilloscope out of my B&W TV?

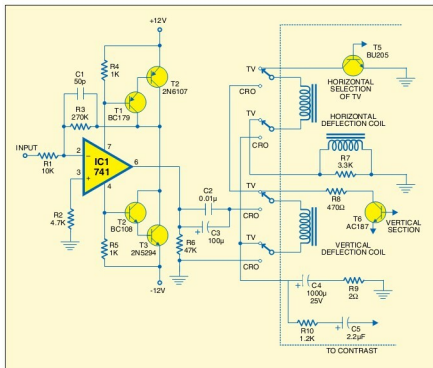
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Through e-mail

A2. Cathode ray oscilloscopes (CROs) are versatile instruments which any professional or hobbyist will like to use for measurements. Unfortunately, the price of a CRO is so high that only professionals can have access to it, while hobbyists and others have to contend with a multimeter to measure voltages.

Every B&W TV is equipped with all the necessary parts of a CRO, so with a few connections you can make a CRO out of a TV without disabling any of its functional features.

In conventional CROs, the deflection of the electron beam is achieved by the electrostatic field between two sets of plates. The input signal is amplified and fed to the vertical deflection plates. The saw-tooth signal is given to the X-plates and serves as the time base.

If you opt for a magnetic field, you



Circuit diagram for conversion of TV into CRO

have to use a coil for deflection. The coil has the property of producing a magnetic field whenever a current passes through it. Here you need the current rather than a high voltage. The resultant interaction between the magnetic fields also reproduces the input signal on the screen, which is the principle of converting a TV into CRO.

Method of conversion. Production of high current is done by power amplifiers. Here approximately $\pm 0.5A$ current is needed for full-scale deflection of the electron beam.

The time base signal, i.e., the conventional saw-tooth signal, is taken from the vertical saw-tooth signal and given to the horizontal deflection coils. This coil is fitted to the neck of the picture tube. The input signal is also power amplified and given to the vertical deflection coils. The complete circuit of the amplifier along with the connections to be made is shown here in the figure.

In the TV itself, there are both vertical saw-tooth and horizontal saw-tooth generators. The horizontal saw-tooth has a frequency of 15.625 Hz, while the vertical saw-tooth has a frequency of 50 Hz,

both of which can be slightly varied.

The impedance of the coil increases with the increase in the applied frequency. This, in turn, affects the current flowing through the coil, thereby decreasing the amount of deflection. Thus a limitation is imposed on the input frequency. So this can be used for demonstration purpose and to see the waveforms in a limited range of frequencies.

Though this is not a calibrated CRO, hobbyists can use it to see voltage waveforms. The cost of the additional circuits (amplifiers) is around Rs 100 only. The power supply required for the circuit is $\pm 12V$ DC.

This circuit can be attached to a colour TV also. In the case of a colour TV, the colour of the waveform can be selected by properly biasing/grounding the appropriate electron gun.

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